

Physics Reference Notes

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Contents

1	Similarity transformations of operators: computing $e^A B e^{-A}$	2
1.1	Motivation	2
1.2	Method A: The Hadamard lemma	2
1.3	Method B: the differential-equation (generator-flow) method	2
1.4	When to prefer which method	3
1.5	Scope: when this machinery fails	3
1.6	References	4

1 Similarity transformations of operators: computing $e^A B e^{-A}$

1.1 Motivation

Similarity transformations $B \mapsto e^A B e^{-A}$ are ubiquitous whenever a Hamiltonian is brought to a simpler (often diagonal) form: Bogoliubov transformations of quadratic Hamiltonians, displacement/squeezing of bosonic modes, polaron (Lang-Firsov) transformations, Schrieffer-Wolff transformations, and passage to the interaction picture or a rotating frame are all instances of this construction. If $A^\dagger = -A$ (generator anti-Hermitian), $U = e^A$ is unitary and the transformation preserves Hermiticity and spectra. This is the case in all examples below.

1.2 Method A: The Hadamard lemma

Hadamard Lemma. *For any two operators A, B ,*

$$e^A B e^{-A} = \sum_{n=0}^{\infty} \frac{1}{n!} \text{ad}_A^n(B), \quad \text{ad}_A(B) \equiv [A, B], \quad \text{ad}_A^n(B) \equiv \underbrace{[A, [A, \dots [A, B] \dots]]}_n. \quad (1)$$

This series is especially helpful whenever the nested commutator yields zero at for a finite n or if the nested commutators eventually reproduce a small, recognizable pattern.

Strategy 1: Reduce to elementary operators. Conjugation is an algebra automorphism: for any X, Y ,

$$e^A (XY) e^{-A} = (e^A X e^{-A}) (e^A Y e^{-A}). \quad (2)$$

It suffices to compute $e^A a e^{-A}$ and $e^A a^\dagger e^{-A}$ once; any polynomial in these follows by substitution and normal-ordering, with no further Hadamard series needed.

Strategy 2: The dagger shortcut. If $A^\dagger = -A$, $U = e^A$ is unitary with $U^\dagger = e^{-A} = U^{-1}$, so $e^A B e^{-A} = U B U^\dagger$; taking the dagger,

$$(e^A B e^{-A})^\dagger = e^A B^\dagger e^{-A}. \quad (3)$$

Once $e^A a e^{-A}$ is known, $e^A a^\dagger e^{-A}$ follows for free by Hermitian conjugation, roughly halving the bookkeeping.

1.3 Method B: the differential-equation (generator-flow) method

Define $X(\theta) \equiv e^{\theta A} X e^{-\theta A}$, $X(0) = X$. Differentiating,

$$\frac{dX(\theta)}{d\theta} = [A, X(\theta)] = \text{ad}_A(X(\theta)). \quad (4)$$

This ODE is exactly equivalent to Method A – the Hadamard series is its formal Taylor solution around $\theta = 0$ – but is often more tractable in practice.

Key fact: closure $\Rightarrow$ finite-dimensional linear ODE. If ad_A maps a finite set of operators $\{X_1, \dots, X_n\}$ into linear combinations of themselves, the operator ODE collapses to a matrix ODE:

$$\frac{d}{d\theta} \vec{X}(\theta) = M \vec{X}(\theta), \quad [A, X_j] = \sum_i M_{ij} X_i, \quad \vec{X}(\theta) = e^{\theta M} \vec{X}(0), \quad (5)$$

reducing the problem to exponentiating an ordinary $n \times n$ matrix.

Why this works for Gaussian generators. If A is at most quadratic in bosonic ladder operators, $[A, a]$ and $[A, a^\dagger]$ are each at most linear, so $\{a, a^\dagger, \mathbb{K}\}$ is automatically closed under ad_A . Quadratic bosonic Hamiltonians generate the (metaplectic representation of the) symplectic group $Sp(2, \mathbb{R})$ acting linearly on phase space.

Worked examples

(a) Squeezing. $A = \frac{\xi}{2}(a^2 - a^{\dagger 2})$: $[A, a] = \xi a^\dagger$, $[A, a^\dagger] = \xi a$, i.e. $M = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Eigenvalues ± 1 (hyperbolic), $M^2 = \mathbb{K}$:

$$a(\xi) = a \cosh \xi + a^\dagger \sinh \xi. \quad (6)$$

(b) Displacement. $A = \alpha(a^\dagger - a)$: $[A, a] = -\alpha \mathbb{K}$ is a c-number, so enlarge the closed set to $\{a, \mathbb{K}\}$. The matrix is nilpotent, giving polynomial growth:

$$a(\alpha) = a - \alpha \mathbb{K}. \quad (7)$$

(c) Phase rotation (interaction picture). $A = i\theta a^\dagger a$: $[a^\dagger a, a] = -a$, closed on $\{a\}$ alone:

$$a(\theta) = e^{-i\theta} a. \quad (8)$$

Classification. Eigenvalues of M purely imaginary $\rightarrow$ oscillatory (elliptic); real nonzero $\rightarrow$ hyperbolic (squeezing); nilpotent/zero $\rightarrow$ polynomial (displacement).

1.4 When to prefer which method

Both methods are mathematically identical. Method A is fast when the pattern is recognizable by eye; Method B is more systematic for anything beyond the simplest single-operator case, especially with mixed or multiple generators to track jointly.

1.5 Scope: when this machinery fails

The closure property requires A to be at most quadratic in a set of canonically (anti)commuting operators. For a genuinely quartic operator such as $H_{\text{int}} = U \sum_i (n_i - \frac{1}{2})(n_{i+1} - \frac{1}{2})$, $\text{ad}_A(H_{\text{int}})$ does not close on a finite set for generic one-body A (e.g. $A \propto J$ or $A \propto T$) – $[J, H_{\text{int}}]$ is generically nonzero and does not reduce to a small closed set of operators. The Hadamard series then cannot be resummed in closed form; one is limited to a perturbative treatment in the strength of A .

1.6 References

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